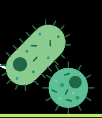
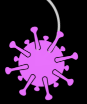


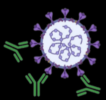
CONTAGION CHRONICLES THE COVID-19 JOURNEY



Caused by the new coronavirus SARS-CoV-2, COVID-19. The main ways that this virus spreads through the respiratory, when COVID-19 presents clinically, symptoms might vary from none at all to a serious respiratory condition that results in death[1].

The immune system serves as the body's line of defense against viruses such as SARS-CoV-2. Antibodies are proteins made by B cells (a subset of white blood cells) that play a major role in this response[2]. Pathogen surfaces contain distinct molecules called antigens, which are recognized and bound to by antibodies[3].

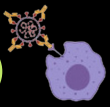
HOW ANTIBODIES WORK



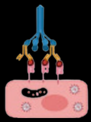
1.Neutralization

By attaching to the surface proteins of viruses, antibodies can neutralize them and stop them from entering and infecting cells[4].

2.Opsonization



In this process, antibodies label pathogens so that other immune cells, like neutrophils and macrophages, can destroy them[4].



3. Complement Activation

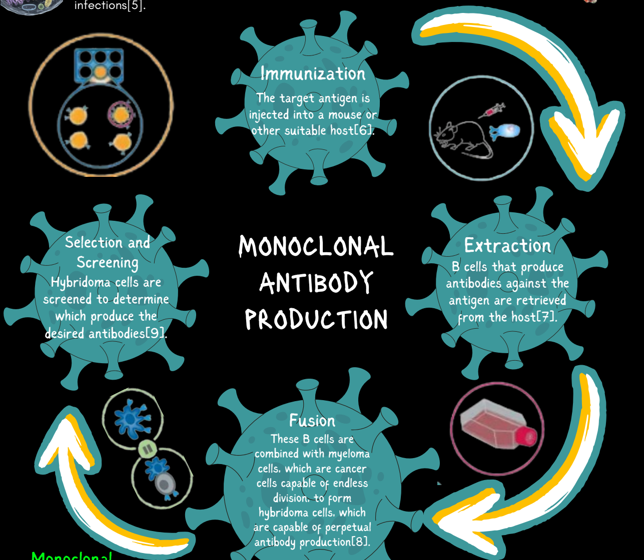
The complement system is a set of proteins that help destroy pathogens and can be activated by antibodies[4].

4.Antibody-Dependent Cellular Cytotoxicity (ADCC)

Through this mechanism, immune cells are drawn to infected cells by antibodies, which then kill the infected cells[4].

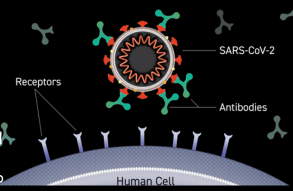
MONOCLONAL ANTIBODIES! A TARGET APPROACH

Monoclonal antibodies (mAbs) are laboratory-created molecules designed to function as substitute antibodies, restoring, enhancing, or mimicking the immune system's attack on infections[5].



Monoclonal Antibodies Against Covid-19

These antibodies typically target the spike protein, which allows the virus to enter human cells. Monoclonal antibodies, by binding to the spike protein, can prevent the virus from adhering to and entering cells, effectively neutralising its capacity to infect[10].



Mechanism of Action

These antibodies can enlist additional immune system components to aid in the removal of the virus and infected cells by binding to the spike protein of SARS-CoV-2. This prevents the virus from adhering to the ACE2 receptors on human cells, which is the first stage in viral entry[11].

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